

Notes: Derivation of ODEs for Atmospheric Propagation Using Lagrangian Approach

Daniel S. Cargill

July 19, 2024

Abstract

1 Introduction

1.1 Helmholtz Equation for Atmospheric Propagation

A monochromatic EM source $E(\tilde{x}, \tilde{y}, \tilde{z}) = A(\tilde{x}, \tilde{y}, \tilde{z}) \exp(-i\omega\tilde{t}) + c.c.$, propagating in a nondispersive static atmosphere follows a stochastic Helmholtz equation

$$\begin{aligned} \nabla^2 \tilde{A} + k^2(n_0 + n(\tilde{x}, \tilde{y}, \tilde{z}))^2 \tilde{A} &= 0 \quad \text{for} \quad -\infty < \tilde{x}, \tilde{y} < \infty, \quad 0 < \tilde{z} < L, \\ \text{where} \quad \tilde{A}(\tilde{x}, \tilde{y}, 0) &= \tilde{A}_i(\tilde{x}, \tilde{y}) \exp(i\Phi(\tilde{x}, \tilde{y})), \end{aligned} \quad (1)$$

and $\tilde{A}_i(\tilde{x}, \tilde{y})$ is of finite support, i.e. $\tilde{A}_i(\tilde{x}, \tilde{y}) = 0$ when $\sqrt{\tilde{x}^2 + \tilde{y}^2} > D_i$. Note, $k = \frac{\omega}{c} = \frac{2\pi}{\lambda}$.

1.2 Atmospheric Statistics

Normally, the stochastic part of the atmosphere is described by the structure function [1]

$$\begin{aligned} \mathbb{E}[(n(\tilde{\mathbf{r}}_1) - n(\tilde{\mathbf{r}}_2))^2] &= C_n^2 \begin{cases} l_0^{-4/3} |\tilde{\mathbf{r}}_1 - \tilde{\mathbf{r}}_2|^2 & 0 < |\tilde{\mathbf{r}}_1 - \tilde{\mathbf{r}}_2| \leq l_0 \\ |\tilde{\mathbf{r}}_1 - \tilde{\mathbf{r}}_2|^{2/3} & l_0 \leq |\tilde{\mathbf{r}}_1 - \tilde{\mathbf{r}}_2| \leq L_0 \end{cases} \\ &= \mathbb{E}[n^2(\tilde{\mathbf{r}}_1)] + \mathbb{E}[n^2(\tilde{\mathbf{r}}_2)] - 2\mathbb{E}[n(\tilde{\mathbf{r}}_1)n(\tilde{\mathbf{r}}_2)] \\ &= 2\sigma_n^2(1 - \text{cor}(n_1, n_2)) \end{aligned} \quad (2)$$

so

$$\text{cor}(n_1, n_2) = 1 - \begin{cases} \frac{C_n^2}{2\sigma_n^2} l_0^{-4/3} |\tilde{\mathbf{r}}_1 - \tilde{\mathbf{r}}_2|^2 & 0 < |\tilde{\mathbf{r}}_1 - \tilde{\mathbf{r}}_2| \leq l_0 \\ \frac{C_n^2}{2\sigma_n^2} |\tilde{\mathbf{r}}_1 - \tilde{\mathbf{r}}_2|^{2/3} & l_0 \leq |\tilde{\mathbf{r}}_1 - \tilde{\mathbf{r}}_2| \leq L_0 \end{cases} \quad (3)$$

If we assume that $|\text{cor}(n_1, n_2)| \ll 1$ for $|\tilde{\mathbf{r}}_1 - \tilde{\mathbf{r}}_2| \gg L_0$, then letting $|\tilde{\mathbf{r}}_1 - \tilde{\mathbf{r}}_2| = C_0 L_0$ for $C_0 > 1$ we can write

$$0 = 1 - C_0^{2/3} \frac{C_n^2 L_0^{2/3}}{2\sigma_n^2}, \quad (4)$$

or

$$\sigma_n^2 = C_0^{2/3} \frac{C_n^2 L_0^{2/3}}{2} \quad (5)$$

and

$$\text{cor}(n_1, n_2) = 1 - \begin{cases} \frac{1}{C_0^{2/3} (L_0 l_0^2)^{2/3}} |\tilde{\mathbf{r}}_1 - \tilde{\mathbf{r}}_2|^2 & 0 < |\tilde{\mathbf{r}}_1 - \tilde{\mathbf{r}}_2| \leq l_0 \\ \frac{1}{C_0^{2/3} L_0^{2/3}} |\tilde{\mathbf{r}}_1 - \tilde{\mathbf{r}}_2|^{2/3} & l_0 \leq |\tilde{\mathbf{r}}_1 - \tilde{\mathbf{r}}_2| \leq L_0 \end{cases} \quad (6)$$

Quantity	Symbol	Characteristic Value(s)
Wavelength	λ	10^{-6} (meters)
Wavenumber	$k = \frac{2\pi}{\lambda}$	10^6 (meters $^{-1}$)
Kolmogorov Inner-scale	l_0	$10^{-2} \leftrightarrow 10^{-3}$ (meters)
Kolmogorov Outer-scale	L_0	$10 \leftrightarrow 10^3$ (meters)
Index Structure Constant	C_n^2	$10^{-20} \leftrightarrow 10^{-12}$ (meters $^{-2/3}$)
Index Variance	$\sigma_n^2 = C_0^{2/3} \frac{C_n^2 L_0^{2/3}}{2}$	$C_0^{2/3} (10^{-18} \leftrightarrow 10^{-10})$
Apperture Diameter	D_i	$1 \leftrightarrow 10^{-2}$ (meters)
Propagation Distance	L	$10^3 \leftrightarrow 10^4$ (meters)

Table 1 – Atmospheric Scales

Quantity	Scale	New Variable
$\tilde{z}$	L_0	$\tilde{z} = L_0 z$
$\tilde{x}$	l_0	$\tilde{x} = l_0 x$
$\tilde{y}$	l_0	$\tilde{y} = l_0 y$
$n(l_0 \tilde{x}, l_0 \tilde{y}, L_0 \tilde{z})$	σ_n	$n(l_0 \tilde{x}, l_0 \tilde{y}, L_0 \tilde{z}) = \sigma_n n(x, y, z)$
$\tilde{A}(l_0 \tilde{x}, l_0 \tilde{y}, L_0 \tilde{z})$	A_0	$\tilde{A}(l_0 \tilde{x}, l_0 \tilde{y}, L_0 \tilde{z}) = A_0 A(x, y, z)$

Table 2 – Dimensional Scales

1.2.1 Nondimensionalization

$$\frac{l_0^2}{L_0^2} \frac{\partial^2 A}{\partial z^2} + \nabla_{\perp}^2 A + l_0^2 k^2 n_0^2 A + 2l_0^2 k^2 n_0 \sigma_n n(x, y, z) A + l_0^2 k^2 \sigma_n^2 n^2(x, y, z) A = 0 \quad (7)$$

for $-\infty < x, y < \infty$, $0 < z < \frac{L}{L_0}$ where $A(x, y, 0) = A_i(x, y) \exp(i\Phi(x, y))$

and $A_i(x, y)$ is of finite support, i.e., $A_i(x, y) = 0$ when $\sqrt{x^2 + y^2} > \frac{D_i}{l_0}$.

1.2.2 Paraxial Approximation

Letting $A = \exp(ikL_0 n_0 z) U(x, y, z)$ gives

$$\frac{l_0^2}{L_0^2} \left(\frac{\partial^2 U}{\partial z^2} + 2ikL_0 n_0 \frac{\partial U}{\partial z} - k^2 L_0^2 n_0^2 U \right) + \nabla_{\perp}^2 U + l_0^2 k^2 n_0^2 U + 2l_0^2 k^2 n_0 \sigma_n n(x, y, z) U + l_0^2 k^2 \sigma_n^2 n^2(x, y, z) U = 0$$

for $-\infty < x, y < \infty$, $0 < z < \frac{L}{L_0}$ where $U(x, y, 0) = U_i(x, y) \exp(i\Phi(x, y))$ (8)

and $U_i(x, y)$ is of finite support, i.e., $U_i(x, y) = 0$ when $\sqrt{x^2 + y^2} > \frac{D_i}{l_0}$.

$$\frac{l_0^2}{L_0^2} \frac{\partial^2 U}{\partial z^2} + 2in_0 \frac{l_0^2 k}{L_0} \frac{\partial U}{\partial z} + \nabla_{\perp}^2 U + 2n_0 l_0^2 k^2 \sigma_n n(x, y, z) U + l_0^2 k^2 \sigma_n^2 n^2(x, y, z) U = 0 \quad (9)$$

for $-\infty < x, y < \infty, \quad 0 < z < \frac{L}{L_0}$ where $U(x, y, 0) = U_i(x, y) \exp(i\Phi(x, y))$

and $U_i(x, y)$ is of finite support, i.e., $U_i(x, y) = 0$ when $\sqrt{x^2 + y^2} > \frac{D_i}{l_0}$.

Ratio	Symbols	Range
Index Variance	$\sigma_n = C_0^{1/3} \frac{C_n L_0^{1/3}}{\sqrt{2}}$	$(10^{-9} \leftrightarrow 10^{-5})$
Inner-scale/Outer-scale	$\epsilon = \frac{l_0}{L_0}$	$10^{-4} \leftrightarrow 10^{-6}$
Inner-scale/Fresnel Length	$\delta = \frac{l_0^2 k}{L_0} = (1 \leftrightarrow 10^2)(10^{-3} \leftrightarrow 10^{-1})$	$10^{-3} \leftrightarrow 10$
Inner-scale/Wavelength	$\gamma = l_0^2 k^2 \sigma_n$	$10^{-3} \leftrightarrow 1$
Inner-scale/Wavelength	$\nu = \gamma \sigma_n$	$10^{-12} \leftrightarrow 10^{-5}$
Propagation Distance/Outer-Scale	$\Lambda = \frac{L}{L_0}$	$1 \leftrightarrow 10$
Aperture Diameter/Inner-Scale	$\Delta_{\perp} = \frac{D_i}{l_0}$	$10^3 \leftrightarrow 10$

Table 3 – Dimensional Scales

$$\epsilon^2 \frac{\partial^2 U}{\partial z^2} + 2in_0 \delta \frac{\partial U}{\partial z} + \nabla_{\perp}^2 U + 2n_0 \gamma n(\mathbf{r}) U + \nu n^2(\mathbf{r}) U = 0 \quad (10)$$

for $-\infty < x, y < \infty, \quad 0 < z < \Lambda$ where $U(x, y, 0) = U_i(x, y) \exp(i\Phi(x, y))$ and $U_i(x, y)$ is of finite support, i.e. $U_i(x, y) = 0$ when $\sqrt{x^2 + y^2} > \Delta_{\perp}$.

$$2in_0 \delta \frac{\partial U}{\partial z} + \nabla_{\perp}^2 U + 2n_0 \gamma n(\mathbf{r}) U = 0 \quad (11)$$

for $-\infty < x, y < \infty, \quad 0 < z < \Lambda$ where $U(x, y, 0) = U_i(x, y) \exp(i\Phi(x, y))$ and $U_i(x, y)$ is of finite support, i.e. $U_i(x, y) = 0$ when $\sqrt{x^2 + y^2} > \Delta_{\perp}$. Also,

$$\text{cor}(n_1, n_2) = 1 - \frac{1}{C_0^{2/3}} \left((z_1 - z_2)^2 + \epsilon |\mathbf{r}_{\perp,1} - \mathbf{r}_{\perp,2}|^2 \right)^{1/3}, \quad \sqrt{\epsilon} \leq \sqrt{(z_1 - z_2)^2 + \epsilon |\mathbf{r}_{\perp,1} - \mathbf{r}_{\perp,2}|^2} \leq 1 \quad (12)$$

$$\text{cor}(n_1, n_2) = 1 - \frac{1}{C_0^{2/3}} \begin{cases} \frac{1}{\epsilon^{2/3}} \left((z_1 - z_2)^2 + \epsilon |\mathbf{r}_{\perp,1} - \mathbf{r}_{\perp,2}|^2 \right)^{1/3} & 0 \leq \sqrt{(z_1 - z_2)^2 + \epsilon |\mathbf{r}_{\perp,1} - \mathbf{r}_{\perp,2}|^2} \leq \sqrt{\epsilon} \\ \left((z_1 - z_2)^2 + \epsilon |\mathbf{r}_{\perp,1} - \mathbf{r}_{\perp,2}|^2 \right)^{1/3} & \sqrt{\epsilon} \leq \sqrt{(z_1 - z_2)^2 + \epsilon |\mathbf{r}_{\perp,1} - \mathbf{r}_{\perp,2}|^2} \leq 1 \end{cases} \quad (13)$$

1.2.3 Numerically

If $z_1 = z_2$

$$\text{cor}(n_1, n_2) = 1 - \frac{1}{C_0^{2/3}} \epsilon^{1/3} |\mathbf{r}_1 - \mathbf{r}_2|^2 \quad 0 \leq |\mathbf{r}_1 - \mathbf{r}_2| \leq 1 \quad (14)$$

$$\text{cor}(n_1, n_2) = 1 - \frac{1}{C_0^{2/3}} \epsilon^{1/3} |\mathbf{r}_1 - \mathbf{r}_2|^{2/3} \quad 1 \leq |\mathbf{r}_1 - \mathbf{r}_2| \leq \frac{1}{\sqrt{\epsilon}} \quad (15)$$

If $1 \geq |z_1 - z_2| \geq \sqrt{\epsilon}$

$$\text{cor}(n_1, n_2) = 1 - \frac{1}{C_0^{2/3}} \left((z_1 - z_2)^2 + \epsilon |\mathbf{r}_1 - \mathbf{r}_2|^2 \right)^{1/3} \quad (16)$$

2 Lagrangian Approach

The Lagrangian approach to atmospheric propagation involves recasting the propagation equation as a functional, where stationary points of the functional are also solutions of the propagation equation. [See Hamilton's Principle](#)

The functional for the paraxial equation above is defined as

$$J(u, \nabla u) = \int_0^\Lambda \iint \mathcal{L}_D(u, \nabla u) dx dy dz \quad (17)$$

where $\mathcal{L}_D(u, \nabla u)$ is referred to as the Lagrangian Density, and is defined as

$$\begin{aligned} \mathcal{L}_D(u, \nabla u) &= in_0\alpha \left(\partial_z(u)\bar{u} - \overline{\partial_z(u)u} \right) - |\nabla_\perp u|^2 + 2n_0\gamma^2 n(\mathbf{r}_\perp, z)|u|^2 \\ &= -2n_0\alpha \operatorname{Im}(\partial_z(u)\bar{u}) - |\nabla_\perp u|^2 + 2n_0\gamma^2 n(\mathbf{r}_\perp, z)|u|^2 \end{aligned} \quad (18)$$

It is fairly easy to show that critical points (points where derivative of the functional is zero) of this functional are also solutions of the propagation equation. Derivatives of a functional are calculated much the same way as derivatives of a vector equation:

$$\begin{aligned} &\lim_{\epsilon \rightarrow 0} \frac{J(u + \epsilon v, \nabla u + \epsilon \nabla v) - J(u, \nabla u)}{\epsilon} \\ &= \lim_{\epsilon \rightarrow 0} \frac{1}{\epsilon} \left[\int_0^\Lambda \iint \mathcal{L}_D(u + \epsilon v, \nabla u + \epsilon \nabla v) dx dy dz - \int_0^\Lambda \iint \mathcal{L}_D(u, \nabla u) dx dy dz \right] \\ &= \int_0^\Lambda \iint \frac{\partial \mathcal{L}_D}{\partial u} v + \frac{\partial \mathcal{L}_D}{\partial(\partial_x(u))} \partial_x(v) + \frac{\partial \mathcal{L}_D}{\partial(\partial_y(u))} \partial_y(v) + \frac{\partial \mathcal{L}_D}{\partial(\partial_z(u))} \partial_z(v) dx dy dz \\ &= \int_0^\Lambda \iint \left[\frac{\partial \mathcal{L}_D}{\partial u} - \frac{\partial}{\partial x} \left(\frac{\partial \mathcal{L}_D}{\partial(\partial_x(u))} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \mathcal{L}_D}{\partial(\partial_y(u))} \right) - \frac{\partial}{\partial z} \left(\frac{\partial \mathcal{L}_D}{\partial(\partial_z(u))} \right) \right] v dx dy dz \\ &\quad + \frac{\partial \mathcal{L}_D}{\partial(\partial_x(u))} v \Big|_{-\infty}^\infty + \frac{\partial \mathcal{L}_D}{\partial(\partial_y(u))} v \Big|_{-\infty}^\infty + \frac{\partial \mathcal{L}_D}{\partial(\partial_z(u))} v \Big|_0^\Lambda. \end{aligned} \quad (19)$$

We can neglect the boundary terms by assuming that $v(0) = v(\Lambda) = \lim_{x \rightarrow \pm\infty} v = \lim_{y \rightarrow \pm\infty} v = 0$. This leaves

$$\frac{\partial J}{\partial u} = \int_0^\Lambda \iint \left[\frac{\partial \mathcal{L}_D}{\partial u} - \frac{\partial}{\partial x} \left(\frac{\partial \mathcal{L}_D}{\partial(\partial_x(u))} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \mathcal{L}_D}{\partial(\partial_y(u))} \right) - \frac{\partial}{\partial z} \left(\frac{\partial \mathcal{L}_D}{\partial(\partial_z(u))} \right) \right] v dx dy dz \quad (20)$$

and thus stationary points are given by u such that

$$\frac{\partial J}{\partial u} = \int_0^\Lambda \iint \left[\frac{\partial \mathcal{L}_D}{\partial u} - \frac{\partial}{\partial x} \left(\frac{\partial \mathcal{L}_D}{\partial(\partial_x(u))} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \mathcal{L}_D}{\partial(\partial_y(u))} \right) - \frac{\partial}{\partial z} \left(\frac{\partial \mathcal{L}_D}{\partial(\partial_z(u))} \right) \right] v dx dy dz = 0. \quad (21)$$

However, the only way this holds for all v is if

$$\frac{\partial \mathcal{L}_D}{\partial u} - \frac{\partial}{\partial x} \left(\frac{\partial \mathcal{L}_D}{\partial(\partial_x(u))} \right) - \frac{\partial}{\partial y} \left(\frac{\partial \mathcal{L}_D}{\partial(\partial_y(u))} \right) - \frac{\partial}{\partial z} \left(\frac{\partial \mathcal{L}_D}{\partial(\partial_z(u))} \right) = 0. \quad (22)$$

This is called the Euler-Lagrange equation(s) and one can verify that inserting the definition for $\mathcal{L}_D$ recovers the paraxial equation discussed above. Thus, stationary points of J are also solutions to the paraxial propagation equation.

2.1 Variational Approximations for Low Dimensional Propagation

One could proceed in a straight forward manor and try to find stationary points of the functional J through numerical algorithms. However, the approach we take here is to use the Lagrangian formulation to find low dimensional (finite dimensional) approximations to the governing equations through variational means. The application of **variational approximations** to PDEs is not a new concept. I was introduced to the methodology while studying the propagation of nonlinear solitary waves (solitons) in optical fiber. The seminal paper for this application was [Anderson, *Variational approach to nonlinear pulse propagation in optical fibers*, (1983)], which introduced the method by following the work done in applying VA methods to soliton propagation in plasmas. The method was further developed for the context of optical fiber in [Anderson, Lisak and Reichel, *Approximate analytical approaches to nonlinear pulse propagation in optical fibers: A comparison*, (1988)].

The approximation procedure begins with postulation of an ansatz (trial solution in analytically form) which has a given function for the solution which allows us to integrate out the x and y dimensions and use the corresponding Euler-Lagrange equations to derive stochastic ODEs for the z evolution.

2.2 Derivation of Gaussian Beam Solution

Consider the unperturbed Helmholtz equation ($n(x, y, z) = 0$)

$$2in_0\alpha\frac{\partial U}{\partial z} + \nabla_{\perp}^2 U = 2in_0\alpha\frac{\partial U}{\partial z} + \frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial U}{\partial r}\right) + \frac{1}{r^2}\frac{\partial U}{\partial \theta^2} = 0 \quad (23)$$

Now, lets look for a radially symmetric solution of the form

$$U \propto \exp\left(-i\left[\frac{kr^2}{2q(z)} + p(z)\right]\right) \quad (24)$$

where $k = n_0\alpha$. The terms become

$$\frac{\partial U}{\partial z} = -i\left[-\frac{kr^2}{2q^2(z)}\dot{q} + \dot{p}\right]U \quad (25)$$

$$\frac{1}{r}\frac{\partial}{\partial r}\left(r\frac{\partial U}{\partial r}\right) = -\left[i\frac{2k}{q} + \frac{k^2}{q^2}r^2\right]U \quad (26)$$

and the equation becomes

$$\begin{aligned} 2ik(-i)\left[-\frac{kr^2}{2q^2}\dot{q} + \dot{p}\right] - \left[i\frac{2k}{q} + \frac{k^2}{q^2}r^2\right] &= \left[-\frac{2k^2r^2}{2q^2}\dot{q} + 2k\dot{p}\right] - \left[i\frac{2k}{q} + \frac{k^2}{q^2}r^2\right] = \\ \left(-\frac{2k^2r^2}{2q^2}\dot{q} - i\frac{2k}{q}\right) + \left(2k\dot{p} - \frac{k^2}{q^2}r^2\right) &= -\frac{k^2r^2}{q^2}(\dot{q} + 1) + 2k\left(\dot{p} - i\frac{1}{q}\right) = 0 \end{aligned} \quad (27)$$

Now, since this equation has to hold for all r , both terms above have to be independently equal to 0.

$$\begin{aligned}\dot{q} + 1 &= 0 \rightarrow q = q(z_0) - (z - z_0) \\ \dot{p} - i\frac{1}{q} &= 0 \rightarrow \dot{p} = i\frac{1}{q(z_0) - (z - z_0)}\end{aligned}\tag{28}$$

For the solution to be of finite energy (L_2), the value of q_0 must be complex. To see this consider if q_0 was real, then $|\exp(-i[\frac{kr^2}{2q(z)} + p(z)])| = |\exp(-ip(z))|$, which doesn't depend on r and hence has infinite energy. In addition, we can absorb any real part of $q(z_0)$ into z_0 . Hence, $q(z_0) = iq_0$ where q_0 is real.

For the p equation, we have

$$\begin{aligned}p(z) &= i \int_{z_0}^z \frac{1}{iq_0 - (z' - z_0)} dz' = -i \int_0^{z-z_0} \frac{1}{z' - iq_0} dz' \\ &= -i \ln(z' - iq_0) \Big|_0^{z-z_0} = -i \ln(z - z_0 - iq_0) + i \ln(-iq_0) \\ &= -i (\ln(z - z_0 - iq_0) - \ln(-iq_0)) \\ &= -i \ln\left(\frac{z - z_0 - iq_0}{-iq_0}\right) = -i \ln\left(1 + i\frac{z - z_0}{q_0}\right)\end{aligned}\tag{29}$$

so

$$\exp(-ip(z)) = \exp\left(-\ln\left(1 + i\frac{z - z_0}{q_0}\right)\right) = \frac{1}{1 + i\frac{z - z_0}{q_0}}\tag{30}$$

Now, we can represent complex numbers $z = a + ib$ as $(a^2 + b^2)^{1/2} \exp(i \arctan(\frac{b}{a}))$ so

$$\exp(-ip(z)) = \frac{1}{\left(1 + \frac{(z - z_0)^2}{q_0^2}\right)^{\frac{1}{2}} \exp\left(i \arctan\left(\frac{z - z_0}{q_0}\right)\right)} = \left(1 + \frac{(z - z_0)^2}{q_0^2}\right)^{-\frac{1}{2}} \exp\left(-i \arctan\left(\frac{z - z_0}{q_0}\right)\right)\tag{31}$$

Finally,

$$\begin{aligned}
U &\propto \left(1 + \frac{(z - z_0)^2}{q_0^2}\right)^{-\frac{1}{2}} \exp\left(i \left[\frac{kr^2}{2((z' - z_0) - iq_0)} \right]\right) \exp\left(-i \arctan\left(\frac{z - z_0}{q_0}\right)\right) \\
&\propto \left(1 + \frac{(z - z_0)^2}{q_0^2}\right)^{-\frac{1}{2}} \exp\left(i \left[\frac{kr^2((z' - z_0) + iq_0)}{2((z' - z_0)^2 + q_0^2)} \right]\right) \exp\left(-i \arctan\left(\frac{z - z_0}{q_0}\right)\right) \\
&\propto \left(1 + \frac{(z - z_0)^2}{q_0^2}\right)^{-\frac{1}{2}} \exp\left(i \left[\frac{kr^2(z' - z_0)}{2((z' - z_0)^2 + q_0^2)} \right]\right) \exp\left(- \left[\frac{kr^2 q_0}{2((z' - z_0)^2 + q_0^2)} \right]\right) \exp\left(-i \arctan\left(\frac{z - z_0}{q_0}\right)\right) \\
&\propto \left(1 + \frac{(z - z_0)^2}{q_0^2}\right)^{-\frac{1}{2}} \exp\left(i \left[\frac{kr^2}{2(z' - z_0) \left(1 + \left(\frac{q_0}{z' - z_0}\right)^2\right)} \right]\right) \exp\left(- \left[\frac{kr^2}{2q_0 \left(1 + \left(\frac{z' - z_0}{q_0}\right)^2\right)} \right]\right) \exp\left(-i \arctan\left(\frac{z - z_0}{q_0}\right)\right)
\end{aligned} \tag{32}$$

Now we can define the spot size and beam waist as

$$w(z) = w_0 \sqrt{1 + \left(\frac{z - z_0}{q_0}\right)^2}, \quad w_0 = \sqrt{q_0} \tag{33}$$

so

$$w^2(z) = q_0 \left(1 + \left(\frac{z - z_0}{q_0}\right)^2\right) = \frac{1}{q_0} (q_0^2 + (z - z_0)^2) \rightarrow q_0 w^2(z) = (q_0^2 + (z - z_0)^2) \tag{34}$$

We can also define radius of curvature as

$$R(z) = (z - z_0) \left[1 + \left(\frac{q_0}{z - z_0}\right)^2\right] \tag{35}$$

so

$$\begin{aligned}
\frac{kr^2(z' - z_0)}{2((z' - z_0)^2 + q_0^2)} &= \frac{kr^2(z' - z_0)}{2(z' - z_0)^2 \left(1 + \frac{q_0^2}{(z' - z_0)^2}\right)} \\
&= \frac{kr^2}{2R(z)}
\end{aligned} \tag{36}$$

Finally,

$$U \propto \frac{w_0}{w(z)} \exp\left(ik \frac{r^2}{2R(z)}\right) \exp\left(-k \frac{r^2}{2w^2(z)}\right) \exp\left(-i \arctan\left(\frac{z - z_0}{q_0}\right)\right) \tag{37}$$

Simplest form is to let $z_0 = 0$, so the spot size is smallest at the origin, and put $k = n_0 \alpha$ and we

get

$$U = A_0 \frac{w_0}{w(z)} \exp\left(in_0\alpha \frac{r^2}{2R(z)}\right) \exp\left(-n_0\alpha \frac{r^2}{2w^2(z)}\right) \exp\left(-i \arctan\left(\frac{z}{q_0}\right)\right) \quad (38)$$

where

$$R(z) = z \left[1 + \left(\frac{q_0}{z}\right)^2\right] \quad w(z) = w_0 \sqrt{1 + \left(\frac{z}{q_0}\right)^2}, \quad w_0 = \sqrt{q_0} \quad (39)$$

or

$$q_0 = w_0^2 \rightarrow R(z) = z \left[1 + \left(\frac{w_0^2}{z}\right)^2\right] \quad (40)$$

and

$$q_0 = w_0^2 \rightarrow w(z) = w_0 \sqrt{1 + \left(\frac{z}{w_0^2}\right)^2} \quad (41)$$

Alternatively, for our purposes we need $z_0 = \Lambda$

$$R(z) = (z - \Lambda) \left[1 + \left(\frac{w_0^2}{(z - \Lambda)}\right)^2\right] \quad (42)$$

and

$$R(0) = -\Lambda \left[1 + \left(\frac{w_0^2}{\Lambda}\right)^2\right] \quad (43)$$

$$w(z) = w_0 \sqrt{1 + \left(\frac{(z - \Lambda)}{w_0^2}\right)^2} \quad (44)$$

and

$$w(0) = w_0 \sqrt{1 + \left(\frac{\Lambda}{w_0^2}\right)^2} \quad (45)$$

So in total...

$$U = A_0 \frac{w_0}{w(z)} \exp\left(in_0\alpha \frac{r^2}{2R(z)}\right) \exp\left(-n_0\alpha \frac{r^2}{2w^2(z)}\right) \exp\left(-i \arctan\left(\frac{z}{w_0^2}\right)\right) \quad (46)$$

$$w(z) = w_0 \sqrt{1 + \left(\frac{z - z_0}{w_0^2}\right)^2} = \frac{w_0}{w_0^2} \sqrt{w_0^4 + (z - z_0)^2} = \frac{1}{w_0} \sqrt{w_0^4 + (z - z_0)^2} \quad (47)$$

$$R(z) = (z - z_0) \left[1 + \left(\frac{w_0^2}{z - z_0}\right)^2 \right] = \frac{1}{z - z_0} [(z - z_0)^2 + w_0^4] \quad (48)$$

But I would rather work with the inverse of these

$$W(z) = \frac{1}{w(z)} = \frac{w_0}{\sqrt{w_0^4 + (z - z_0)^2}} \quad (49)$$

$$F(z) = \frac{1}{R(z)} = \frac{z - z_0}{(z - z_0)^2 + w_0^4} \quad (50)$$

with solution

$$U = A_0 w_0 W(z) \exp\left(i \frac{n_0 \alpha}{2} F(z) r^2\right) \exp\left(-\frac{n_0 \alpha}{2} W^2(z) r^2\right) \exp\left(-i \arctan\left(\frac{z}{w_0^2}\right)\right) \quad (51)$$

2.2.1 Normalization

$$|U|^2 = A_0^2 \frac{w_0^2}{w^2(z)} \exp\left(-n_0 \alpha \frac{r^2}{w^2(z)}\right) \quad (52)$$

so

$$\iint |U|^2 dx dy = 2\pi A_0^2 \frac{w_0^2}{w^2(z)} \int_0^\infty \exp\left(-n_0 \alpha \frac{r^2}{w^2(z)}\right) r dr \quad (53)$$

letting $r = \frac{w(z)}{\sqrt{n_0 \alpha}} x$ so $dr = \frac{w(z)}{\sqrt{n_0 \alpha}} dx$

$$2\pi A_0^2 \frac{w_0^2}{w^2(z)} \int_0^\infty \exp(-r^2) \frac{w(z)}{\sqrt{n_0 \alpha}} x \frac{w(z)}{\sqrt{n_0 \alpha}} dx \quad (54)$$

$$2\pi A_0^2 \frac{w_0^2}{n_0 \alpha} \int_0^\infty \exp(-x^2) x dx = \pi A_0^2 \frac{w_0^2}{n_0 \alpha} = 1 \quad (55)$$

$$A_0 = \sqrt{\frac{n_0 \alpha}{\pi w_0^2}} = \sqrt{\frac{n_0 \alpha}{\pi}} \frac{1}{w_0} \quad (56)$$

$$W(z) = \frac{1}{w(z)} = \frac{w_0}{\sqrt{w_0^4 + (z - z_0)^2}} \quad (57)$$

$$F(z) = \frac{1}{R(z)} = \frac{z - z_0}{(z - z_0)^2 + w_0^4} \quad (58)$$

with solution

$$U = \sqrt{\frac{n_0 \alpha}{\pi}} W(z) \exp\left(i \frac{n_0 \alpha}{2} F(z) r^2\right) \exp\left(-\frac{n_0 \alpha}{2} W^2(z) r^2\right) \exp\left(-i \arctan\left(\frac{z}{w_0^2}\right)\right) \quad (59)$$

3 Gaussian Ansatz: General

We begin by using a Gaussian ansatz:

$$u(\mathbf{r}_\perp, \mathbf{p}(z)) = I(\mathbf{p}) \exp(-(\Theta(\mathbf{r}_\perp, \mathbf{p}) + i\Phi(\mathbf{r}_\perp, \mathbf{p}))) \quad (60)$$

Now, lets calculate each term in the Lagrangian density:

$$\begin{aligned} \partial_z(u) &= \left[\frac{\nabla \mathbf{p} I \cdot \dot{\mathbf{p}}}{I} - \nabla \mathbf{p} \Theta \cdot \dot{\mathbf{p}} - i \nabla \mathbf{p} \Phi \cdot \dot{\mathbf{p}} \right] u \\ \partial_z(u) \bar{u} &= \left[\frac{\nabla \mathbf{p} I \cdot \dot{\mathbf{p}}}{I} - \nabla \mathbf{p} \Theta \cdot \dot{\mathbf{p}} - i \nabla \mathbf{p} \Phi \cdot \dot{\mathbf{p}} \right] |u|^2 \\ -2n_0 \alpha \text{Im} [\partial_z(u) \bar{u}] &= 2n_0 \alpha (\nabla \mathbf{p} \Phi \cdot \dot{\mathbf{p}}) |u|^2 \\ &= 2n_0 \alpha (\nabla \mathbf{p} \Phi \cdot \dot{\mathbf{p}}) I^2(\mathbf{p}) \exp(-2\Theta(\mathbf{r}_\perp, \mathbf{p})) \end{aligned} \quad (61)$$

$$\begin{aligned} \nabla_\perp u &= -(\nabla_\perp \Theta(\mathbf{r}_\perp, \mathbf{p}) + i \nabla_\perp \Phi(\mathbf{r}_\perp, \mathbf{p})) u \\ -|\nabla_\perp u|^2 &= -(|\nabla_\perp \Theta(\mathbf{r}_\perp, \mathbf{p})|^2 + |\nabla_\perp \Phi(\mathbf{r}_\perp, \mathbf{p})|^2) |u|^2 \\ &= -(|\nabla_\perp \Theta(\mathbf{r}_\perp, \mathbf{p})|^2 + |\nabla_\perp \Phi(\mathbf{r}_\perp, \mathbf{p})|^2) I^2(\mathbf{p}) \exp(-2\Theta(\mathbf{r}_\perp, \mathbf{p})) \end{aligned} \quad (62)$$

$$|u|^2 = I^2(\mathbf{p}) \exp(-2\Theta(\mathbf{r}_\perp, \mathbf{p})) \quad (63)$$

$$\begin{aligned} \mathcal{L}_D(u, \nabla u) &= -2n_0 \alpha \text{Im} (\partial_z(u) \bar{u}) - |\nabla_\perp u|^2 + 2n_0 \gamma^2 n(\mathbf{r}_\perp, z) |u|^2 \\ &= (2n_0 \alpha \nabla \mathbf{p} \Phi \cdot \dot{\mathbf{p}} - (|\nabla_\perp \Theta(\mathbf{r}_\perp, \mathbf{p})|^2 + |\nabla_\perp \Phi(\mathbf{r}_\perp, \mathbf{p})|^2) + 2n_0 \gamma^2 n(\mathbf{r}_\perp, z)) I^2(\mathbf{p}) \exp(-2\Theta(\mathbf{r}_\perp, \mathbf{p})) \end{aligned} \quad (64)$$

3.1 Tilted Gaussian

$$\mathbf{p} = [A, W_x, W_y; T_x, T_y, X, Y; F_x, F_y, P] \quad (65)$$

$$\begin{aligned} I(\mathbf{p}) &= \frac{A \sqrt{W_x W_y}}{\sqrt{\pi}} \\ \Theta(\mathbf{r}_\perp, \mathbf{p}) &= \frac{1}{2} (W_x^2 (x - X)^2 + W_y^2 (y - Y)^2) \\ \Phi(\mathbf{r}_\perp, \mathbf{p}) &= P + T_x (x - X) + T_y (y - Y) + F_x (x - X)^2 + F_y (y - Y)^2 \end{aligned} \quad (66)$$

$$\mathbf{p} = [A, W_x, W_y; T_x, T_y, X, Y; F_x, F_y, P] \quad (67)$$

$$\begin{aligned}\nabla_{\mathbf{p}}\Phi \cdot \dot{\mathbf{p}} &= \dot{P} + \dot{T}_x(x - X) + \dot{T}_y(y - Y) + \dot{F}_x(x - X)^2 + \dot{F}_y(y - Y)^2 \\ &\quad - \dot{X}(T_x + 2F_x(x - X)) - \dot{Y}(T_y + 2F_y(y - Y))\end{aligned}\quad (68)$$

$$\begin{aligned}\nabla_{\perp}\Theta(\mathbf{r}_{\perp}, \mathbf{p}) &= [W_x^2(x - X), W_y^2(y - Y)] \\ |\nabla_{\perp}\Theta(\mathbf{r}_{\perp}, \mathbf{p})|^2 &= W_x^4(x - X)^2 + W_y^4(y - Y)^2\end{aligned}\quad (69)$$

$$\begin{aligned}\nabla_{\perp}\Phi(\mathbf{r}_{\perp}, \mathbf{p}) &= [T_x + 2F_x(x - X), T_y + 2F_y(y - Y)] \\ |\nabla_{\perp}\Phi(\mathbf{r}_{\perp}, \mathbf{p})|^2 &= (T_x + 2F_x(x - X))^2 + (T_y + 2F_y(y - Y))^2\end{aligned}\quad (70)$$

$$\begin{aligned}\mathcal{L}_D(u, \nabla u) &= 2n_0\alpha \left(\dot{P} + \dot{T}_x(x - X) + \dot{T}_y(y - Y) + \dot{F}_x(x - X)^2 + \dot{F}_y(y - Y)^2 \right) \frac{A^2 W_x W_y}{\pi} \exp(- (W_x^2(x - X)^2 + W_y^2(y - Y)^2)) \\ &\quad - \dot{X}(T_x + 2F_x(x - X)) \frac{A^2 W_x W_y}{\pi} \exp(- (W_x^2(x - X)^2 + W_y^2(y - Y)^2)) \\ &\quad - \dot{Y}(T_y + 2F_y(y - Y)) \frac{A^2 W_x W_y}{\pi} \exp(- (W_x^2(x - X)^2 + W_y^2(y - Y)^2)) \\ &\quad - (W_x^4(x - X)^2 + W_y^4(y - Y)^2) \frac{A^2 W_x W_y}{\pi} \exp(- (W_x^2(x - X)^2 + W_y^2(y - Y)^2)) \\ &\quad - (T_x + 2F_x(x - X))^2 \frac{A^2 W_x W_y}{\pi} \exp(- (W_x^2(x - X)^2 + W_y^2(y - Y)^2)) \\ &\quad - (T_y + 2F_y(y - Y))^2 \frac{A^2 W_x W_y}{\pi} \exp(- (W_x^2(x - X)^2 + W_y^2(y - Y)^2)) \\ &\quad + 2n_0\gamma^2 n(\mathbf{r}_{\perp}, z) \frac{A^2 W_x W_y}{\pi} \exp(- (W_x^2(x - X)^2 + W_y^2(y - Y)^2))\end{aligned}\quad (71)$$

$$\begin{aligned}
\mathcal{F}_D(\mathbf{p}, \dot{\mathbf{p}}) = & 2n_0\alpha \frac{A^2 W_x W_y}{\pi} \left(\dot{P} I_0 + \dot{T}_x I_{1,x} + \dot{T}_y I_{1,y} + \dot{F}_x I_{2,x} + \dot{F}_y I_{2,y} \right) \\
& - \dot{X} \frac{A^2 W_x W_y}{\pi} (T_x I_0 + 2F_x I_{1,x}) \\
& - \dot{Y} \frac{A^2 W_x W_y}{\pi} (T_y I_0 + 2F_y I_{1,y}) \\
& - \frac{A^2 W_x W_y}{\pi} (W_x^4 I_{2,x} + W_y^4 I_{2,y}) \\
& - \frac{A^2 W_x W_y}{\pi} (T_x^2 I_0 + 2T_x F_x I_{1,x} + 4F_x^2 I_{2,x}) \\
& - \frac{A^2 W_x W_y}{\pi} (T_y^2 I_0 + 2T_y F_y I_{1,y} + 4F_y^2 I_{2,y}) \\
& + 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{A^2 W_x W_y}{\pi} \exp(- (W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \right\rangle
\end{aligned} \tag{72}$$

$$\begin{aligned}
I_0 &= \iint_{-\infty}^{\infty} \exp(- (W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \, dx \, dy = \frac{\pi}{W_x W_y} \\
I_{1,x} &= \iint_{-\infty}^{\infty} (x-X) \exp(- (W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \, dx \, dy = 0 \\
I_{1,y} &= \iint_{-\infty}^{\infty} (y-Y) \exp(- (W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \, dx \, dy = 0 \\
I_{2,x} &= \iint_{-\infty}^{\infty} (x-X)^2 \exp(- (W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \, dx \, dy = \frac{\pi}{2W_x^3 W_y} \\
I_{2,y} &= \iint_{-\infty}^{\infty} (y-Y)^2 \exp(- (W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \, dx \, dy = \frac{\pi}{2W_y^3 W_x}
\end{aligned} \tag{73}$$

$$\begin{aligned}
\mathcal{F}_D(\mathbf{p}, \dot{\mathbf{p}}) = & 2n_0\alpha \frac{A^2 W_x W_y}{\pi} \left(\dot{P} \frac{\pi}{W_x W_y} + \dot{F}_x \frac{\pi}{2W_x^3 W_y} + \dot{F}_y \frac{\pi}{2W_y^3 W_x} \right) \\
& - \dot{X} \frac{A^2 W_x W_y}{\pi} T_x \frac{\pi}{W_x W_y} \\
& - \dot{Y} \frac{A^2 W_x W_y}{\pi} T_y \frac{\pi}{W_x W_y} \\
& - \frac{A^2 W_x W_y}{\pi} \left(W_x^4 \frac{\pi}{2W_x^3 W_y} + W_y^4 \frac{\pi}{2W_y^3 W_x} \right) \\
& - \frac{A^2 W_x W_y}{\pi} \left(T_x^2 \frac{\pi}{W_x W_y} + 4F_x^2 \frac{\pi}{2W_x^3 W_y} \right) \\
& - \frac{A^2 W_x W_y}{\pi} \left(T_y^2 \frac{\pi}{W_x W_y} + 4F_y^2 \frac{\pi}{2W_y^3 W_x} \right) \\
& + 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{A^2 W_x W_y}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \right\rangle
\end{aligned} \tag{74}$$

$$\begin{aligned}
\mathcal{F}_D(\mathbf{p}, \dot{\mathbf{p}}) = & 2n_0\alpha A^2 \left(\dot{P} + \frac{\dot{F}_x}{2W_x^2} + \frac{\dot{F}_y}{2W_y^2} \right) - A^2 \dot{X} T_x - A^2 \dot{Y} T_y - A^2 \left(\frac{W_x^2}{2} + \frac{W_y^2}{2} \right) \\
& - A^2 \left(T_x^2 + 2 \frac{F_x^2}{W_x^2} \right) - A^2 \left(T_y^2 + 2 \frac{F_y^2}{W_y^2} \right) \\
& + 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{A^2 W_x W_y}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \right\rangle
\end{aligned} \tag{75}$$

$$\begin{aligned}
\mathcal{F}_D(\mathbf{p}, \dot{\mathbf{p}}) = & 2n_0\alpha A^2 \left(\dot{P} + \frac{\dot{F}_x}{2W_x^2} + \frac{\dot{F}_y}{2W_y^2} \right) - A^2 (\dot{X} T_x + \dot{Y} T_y) - \frac{A^2}{2} (W_x^2 + W_y^2) \\
& - A^2 (T_x^2 + T_y^2) - 2A^2 \left(\frac{F_x^2}{W_x^2} + \frac{F_y^2}{W_y^2} \right) \\
& + 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{A^2 W_x W_y}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \right\rangle
\end{aligned} \tag{76}$$

3.2 Extracting Parameters

$$\begin{aligned}
\int |u|^2 dx dy = & \frac{A^2 W_x W_y}{\pi} \int \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) dx dy = \\
& \frac{A^2}{\pi} \int \exp(-(\hat{x}^2 + \hat{y}^2)) d\hat{x} d\hat{y} = A^2
\end{aligned} \tag{77}$$

$$\begin{aligned} \int x|u|^2 dx dy &= \frac{A^2 W_x W_y}{\pi} \int x \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) dx dy = \\ &= \frac{A^2}{\sqrt{\pi}} \int \left(\frac{\hat{x}}{W_x} + X\right) \exp(-\hat{x}^2) d\hat{x} = A^2 = A^2 X \end{aligned} \quad (78)$$

$$\begin{aligned} \int (x-X)^2 |u|^2 dx dy &= \frac{A^2 W_x W_y}{\pi} \int (x-X)^2 \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) dx dy = \\ &= \frac{A^2}{\sqrt{\pi}} \int \frac{\hat{x}^2}{W_x^2} \exp(-\hat{x}^2) d\hat{x} = \frac{A^2}{2W_x^2} \end{aligned} \quad (79)$$

3.3 Euler-Lagrange equation(s)

Now, we can apply the Euler-Lagrange equations to the reduced density $\mathcal{F}_D(\mathbf{p}, \dot{\mathbf{p}})$, i.e.

$$\frac{d\mathcal{F}_D}{dp_i} - \frac{d}{dz} \left(\frac{\partial \mathcal{F}_D}{\partial \dot{p}_i} \right) = 0 \quad (80)$$

for $p_i = A, W_x, W_y; T_x, T_y, X, Y; F_x, F_y, P$. So lets calculate the terms on the left side of each:

$$\begin{aligned} \frac{dF_D}{dA} &= 2n_0\alpha(2A) \left(\dot{P} + \frac{\dot{F}_x}{2W_x^2} + \frac{\dot{F}_y}{2W_y^2} \right) - 2A \left(\dot{X}T_x + \dot{Y}T_y \right) - A(W_x^2 + W_y^2) \\ &\quad - 2A(T_x^2 + T_y^2) - 4A \left(\frac{F_x^2}{W_x^2} + \frac{F_y^2}{W_y^2} \right) + 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial A} \right\rangle \\ \frac{d}{dz} \left(\frac{\partial F_D}{\partial \dot{A}} \right) &= \frac{d}{dz}(0) = 0 \end{aligned} \quad (81)$$

$$\begin{aligned} \frac{dF_D}{dW_x} &= 2n_0\alpha A^2 \left(-\frac{\dot{F}_x}{W_x^3} \right) - A^2 W_x + 4A^2 \frac{F_x^2}{W_x^3} + 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_x} \right\rangle \\ \frac{d}{dz} \left(\frac{\partial F_D}{\partial \dot{W}_x} \right) &= \frac{d}{dz}(0) = 0 \end{aligned} \quad (82)$$

$$\begin{aligned} \frac{dF_D}{dW_y} &= 2n_0\alpha A^2 \left(-\frac{\dot{F}_y}{W_y^3} \right) - A^2 W_y + 4A^2 \frac{F_y^2}{W_y^3} + 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_y} \right\rangle \\ \frac{d}{dz} \left(\frac{\partial F_D}{\partial \dot{W}_y} \right) &= \frac{d}{dz}(0) = 0 \end{aligned} \quad (83)$$

$$\begin{aligned}\frac{dF_D}{dT_x} &= -A^2\dot{X} - 2A^2T_x + 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial|u|^2}{\partial T_x} \right\rangle \\ \frac{d}{dz} \left(\frac{\partial F_D}{\partial \dot{T}_x} \right) &= \frac{d}{dz}(0) = 0\end{aligned}\tag{84}$$

$$\begin{aligned}\frac{dF_D}{dT_y} &= -A^2\dot{Y} - 2A^2T_y + 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial|u|^2}{\partial T_y} \right\rangle \\ \frac{d}{dz} \left(\frac{\partial F_D}{\partial \dot{T}_y} \right) &= \frac{d}{dz}(0) = 0\end{aligned}\tag{85}$$

$$\begin{aligned}\frac{dF_D}{dX} &= 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial|u|^2}{\partial X} \right\rangle \\ \frac{d}{dz} \left(\frac{\partial F_D}{\partial \dot{X}} \right) &= \frac{d}{dz}(-A^2T_x)\end{aligned}\tag{86}$$

$$\begin{aligned}\frac{dF_D}{dY} &= 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial|u|^2}{\partial Y} \right\rangle \\ \frac{d}{dz} \left(\frac{\partial F_D}{\partial \dot{Y}} \right) &= \frac{d}{dz}(-A^2T_y)\end{aligned}\tag{87}$$

$$\begin{aligned}\frac{dF_D}{dF_x} &= -4A^2 \frac{F_x}{W_x^2} + 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial|u|^2}{\partial F_x} \right\rangle \\ \frac{d}{dz} \left(\frac{\partial F_D}{\partial \dot{F}_x} \right) &= 2n_0\alpha \frac{d}{dz} \left(A^2 \frac{1}{2W_x^2} \right)\end{aligned}\tag{88}$$

$$\begin{aligned}\frac{dF_D}{dF_y} &= -4A^2 \frac{F_y}{W_y^2} + 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial|u|^2}{\partial F_y} \right\rangle \\ \frac{d}{dz} \left(\frac{\partial F_D}{\partial \dot{F}_y} \right) &= 2n_0\alpha \frac{d}{dz} \left(A^2 \frac{1}{2W_y^2} \right)\end{aligned}\tag{89}$$

$$\begin{aligned}\frac{dF_D}{dP} &= 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial|u|^2}{\partial P} \right\rangle \\ \frac{d}{dz} \left(\frac{\partial F_D}{\partial \dot{P}} \right) &= 2n_0\alpha \frac{d}{dz}(A^2)\end{aligned}\tag{90}$$

3.3.1 Equations in P

Since $|u|^2$ is only dependent on A, W , not R, P , the last of these equations says

$$\frac{d}{dz} (2n_0\alpha A^2) = 0, \quad (91)$$

or that

$$A(z) = A(0) \quad (92)$$

3.3.2 Equations in F_x, F_y

$$\begin{aligned} -4A^2 \frac{F_x}{W_x^2} + 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial F_x} \right\rangle &= 2n_0\alpha \frac{d}{dz} \left(A^2 \frac{1}{2W_x^2} \right) = -2n_0\alpha A^2 \frac{\dot{W}_x}{W_x^3} \\ A^2 \frac{\dot{W}_x}{W_x^3} &= \frac{4}{2n_0\alpha} A^2 \frac{F_x}{W_x^2} - \frac{2n_0\gamma^2}{2n_0\alpha} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial F_x} \right\rangle \\ \dot{W}_x &= \frac{2}{n_0\alpha} F_x W_x - \frac{\gamma^2}{\alpha} \frac{W_x^3}{A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial F_x} \right\rangle \\ \dot{W}_x &= \frac{2}{n_0\alpha} F_x W_x \end{aligned} \quad (93)$$

$$\begin{aligned} -4A^2 \frac{F_y}{W_y^2} + 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial F_y} \right\rangle &= 2n_0\alpha \frac{d}{dz} \left(A^2 \frac{1}{2W_y^2} \right) = -2n_0\alpha A^2 \frac{\dot{W}_y}{2W_y^3} \\ \dot{W}_y &= \frac{2}{n_0\alpha} F_y W_y - \frac{\gamma^2}{\alpha} \frac{W_y^3}{A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial F_y} \right\rangle \\ \dot{W}_y &= \frac{2}{n_0\alpha} F_y W_y \end{aligned} \quad (94)$$

3.3.3 Equations in X, Y

$$\begin{aligned} 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial X} \right\rangle &= \frac{d}{dz} (-A^2 T_x) = -A^2 \dot{T}_x \\ \dot{T}_x &= -2n_0\gamma^2 \frac{1}{A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial X} \right\rangle \end{aligned} \quad (95)$$

$$\begin{aligned} 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial Y} \right\rangle &= \frac{d}{dz} (-A^2 T_y) \\ \dot{T}_y &= -2n_0\gamma^2 \frac{1}{A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial Y} \right\rangle \end{aligned} \quad (96)$$

3.3.4 Equations in T_x, T_y

$$\begin{aligned}
-A^2 \dot{X} - 2A^2 T_x + 2n_0 \gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial T_x} \right\rangle &= 0 \\
\dot{X} &= -2T_x + 2n_0 \gamma^2 \frac{1}{A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial T_x} \right\rangle \\
\dot{X} &= -2T_x
\end{aligned} \tag{97}$$

$$\begin{aligned}
-A^2 \dot{Y} - 2A^2 T_y + 2n_0 \gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial T_y} \right\rangle &= 0 \\
\dot{Y} &= -2T_y + 2n_0 \gamma^2 \frac{1}{A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial T_y} \right\rangle \\
\dot{Y} &= -2T_y
\end{aligned} \tag{98}$$

3.3.5 Equations in W_x, W_y

$$\begin{aligned}
2n_0 \alpha A^2 \left(-\frac{\dot{F}_x}{W_x^3} \right) - A^2 W_x + 4A^2 \frac{F_x^2}{W_x^3} + 2n_0 \gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_x} \right\rangle &= 0 \\
2n_0 \alpha A^2 \frac{\dot{F}_x}{W_x^3} &= -A^2 W_x + 4A^2 \frac{F_x^2}{W_x^3} + 2n_0 \gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_x} \right\rangle \\
\frac{\dot{F}_x}{W_x^3} &= -\frac{A^2 W_x}{2n_0 \alpha A^2} + \frac{4A^2}{2n_0 \alpha A^2} \frac{F_x^2}{W_x^3} + \frac{2n_0 \gamma^2}{2n_0 \alpha A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_x} \right\rangle \\
\dot{F}_x &= -\frac{W_x^4}{2n_0 \alpha} + \frac{2W_x^3}{n_0 \alpha} \frac{F_x^2}{W_x^3} + \frac{\gamma^2 W_x^3}{\alpha A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_x} \right\rangle \\
\dot{F}_x &= -\frac{1}{2n_0 \alpha} W_x^4 + \frac{2}{n_0 \alpha} F_x^2 + \frac{\gamma^2 W_x^3}{\alpha A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_x} \right\rangle
\end{aligned} \tag{99}$$

$$\begin{aligned}
2n_0 \alpha A^2 \left(-\frac{\dot{F}_y}{W_y^3} \right) - A^2 W_y + 4A^2 \frac{F_y^2}{W_y^3} + 2n_0 \gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_y} \right\rangle &= 0 \\
\dot{F}_y &= -\frac{1}{2n_0 \alpha} W_y^4 + \frac{2}{n_0 \alpha} F_y^2 + \frac{\gamma^2 W_y^3}{\alpha A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_y} \right\rangle
\end{aligned} \tag{100}$$

3.3.6 Equations in A

$$\begin{aligned}
& 2n_0\alpha(2A) \left(\dot{P} + \frac{\dot{F}_x}{2W_x^2} + \frac{\dot{F}_y}{2W_y^2} \right) - 2A \left(\dot{X}T_x + \dot{Y}T_y \right) - A(W_x^2 + W_y^2) \\
& - 2A(T_x^2 + T_y^2) - 4A \left(\frac{F_x^2}{W_x^2} + \frac{F_y^2}{W_y^2} \right) + 2n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{\partial|u|^2}{\partial A} \right\rangle = 0 \\
& \left(\dot{P} + \frac{\dot{F}_x}{2W_x^2} + \frac{\dot{F}_y}{2W_y^2} \right) = \frac{2A}{2n_0\alpha(2A)} \left(\dot{X}T_x + \dot{Y}T_y \right) + \frac{A}{2n_0\alpha(2A)} (W_x^2 + W_y^2) \\
& + \frac{2A}{2n_0\alpha(2A)} (T_x^2 + T_y^2) + \frac{4A}{2n_0\alpha(2A)} \left(\frac{F_x^2}{W_x^2} + \frac{F_y^2}{W_y^2} \right) - \frac{2n_0\gamma^2}{2n_0\alpha(2A)} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial|u|^2}{\partial A} \right\rangle \\
& \left(\dot{P} + \frac{\dot{F}_x}{2W_x^2} + \frac{\dot{F}_y}{2W_y^2} \right) = \frac{1}{2n_0\alpha} \left(\dot{X}T_x + \dot{Y}T_y \right) + \frac{1}{4n_0\alpha} (W_x^2 + W_y^2) \\
& + \frac{1}{2n_0\alpha} (T_x^2 + T_y^2) + \frac{1}{n_0\alpha} \left(\frac{F_x^2}{W_x^2} + \frac{F_y^2}{W_y^2} \right) - \frac{\gamma^2}{2\alpha A} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial|u|^2}{\partial A} \right\rangle
\end{aligned} \tag{101}$$

$$\begin{aligned}
\dot{P} &= -\frac{\dot{F}_x}{2W_x^2} - \frac{\dot{F}_y}{2W_y^2} + \frac{1}{2n_0\alpha} \left(\dot{X}T_x + \dot{Y}T_y \right) + \frac{1}{4n_0\alpha} (W_x^2 + W_y^2) \\
&+ \frac{1}{2n_0\alpha} (T_x^2 + T_y^2) + \frac{1}{n_0\alpha} \left(\frac{F_x^2}{W_x^2} + \frac{F_y^2}{W_y^2} \right) - \frac{\gamma^2}{2\alpha A} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial|u|^2}{\partial A} \right\rangle
\end{aligned} \tag{102}$$

$$\begin{aligned}
\dot{P} = & -\frac{1}{2W_x^2} \left(-\frac{1}{2n_0\alpha} W_x^4 + \frac{2}{n_0\alpha} F_x^2 + \frac{\gamma^2 W_x^3}{\alpha A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_x} \right\rangle \right) \\
& -\frac{1}{2W_y^2} \left(-\frac{1}{2n_0\alpha} W_y^4 + \frac{2}{n_0\alpha} F_y^2 + \frac{\gamma^2 W_y^3}{\alpha A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_y} \right\rangle \right) \\
& + \frac{1}{2n_0\alpha} T_x (-2T_x) \\
& + \frac{1}{2n_0\alpha} T_y (-2T_y) \\
& + \frac{1}{4n_0\alpha} (W_x^2 + W_y^2) \\
& + \frac{1}{2n_0\alpha} (T_x^2 + T_y^2) \\
& + \frac{1}{n_0\alpha} \left(\frac{F_x^2}{W_x^2} + \frac{F_y^2}{W_y^2} \right) \\
& - \frac{\gamma^2}{2\alpha A} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial A} \right\rangle \\
= & \left(\frac{1}{4n_0\alpha} W_x^2 - \frac{1}{n_0\alpha} \frac{F_x^2}{W_x^2} - \frac{\gamma^2 W_x}{2\alpha A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_x} \right\rangle \right) \\
& \left(\frac{1}{4n_0\alpha} W_y^2 - \frac{1}{n_0\alpha} \frac{F_y^2}{W_y^2} - \frac{\gamma^2 W_y}{2\alpha A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_y} \right\rangle \right) \\
& + \frac{1}{4n_0\alpha} (W_x^2 + W_y^2) \\
& + \frac{1}{n_0\alpha} \left(\frac{F_x^2}{W_x^2} + \frac{F_y^2}{W_y^2} \right) \\
& - \frac{\gamma^2}{2\alpha A} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial A} \right\rangle
\end{aligned} \tag{103}$$

$$\begin{aligned}
\dot{P} = & \frac{1}{2n_0\alpha} (W_x^2 + W_y^2) - \frac{\gamma^2 W_x}{2\alpha A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_x} \right\rangle - \frac{\gamma^2 W_y}{2\alpha A^2} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial W_y} \right\rangle \\
& - \frac{\gamma^2}{2\alpha A} \left\langle n(\mathbf{r}_\perp, z), \frac{\partial |u|^2}{\partial A} \right\rangle
\end{aligned} \tag{104}$$

$$\dot{P} = \frac{1}{2n_0\alpha} (W_x^2 + W_y^2) - \frac{\gamma^2}{2\alpha} \left\langle n(\mathbf{r}_\perp, z), \frac{W_x}{A^2} \frac{\partial |u|^2}{\partial W_x} + \frac{W_y}{A^2} \frac{\partial |u|^2}{\partial W_y} + \frac{1}{A} \frac{\partial |u|^2}{\partial A} \right\rangle \tag{105}$$

3.4 Equations

$$A(z) = A(0)$$

$$\dot{P} = \frac{1}{2n_0\alpha} (W_x^2 + W_y^2) - \frac{\gamma^2}{\alpha} \left\langle n(\mathbf{r}_\perp, z), \frac{W_x}{2A^2} \frac{\partial |u|^2}{\partial W_x} + \frac{W_y}{2A^2} \frac{\partial |u|^2}{\partial W_y} + \frac{1}{2A} \frac{\partial |u|^2}{\partial A} \right\rangle \quad (106)$$

$$\begin{aligned} \dot{X} &= -2T_x \\ \dot{Y} &= -2T_y \\ \dot{T}_x &= -n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{2}{A^2} \frac{\partial |u|^2}{\partial X} \right\rangle \\ \dot{T}_y &= -n_0\gamma^2 \left\langle n(\mathbf{r}_\perp, z), \frac{2}{A^2} \frac{\partial |u|^2}{\partial Y} \right\rangle \end{aligned} \quad (107)$$

$$\begin{aligned} \dot{W}_x &= 4F_x W_x \\ \dot{W}_y &= 4F_y W_y \\ \dot{F}_x &= -\frac{1}{2n_0\alpha} W_x^4 + \frac{2}{n_0\alpha} F_x^2 + \frac{\gamma^2}{\alpha} \left\langle n(\mathbf{r}_\perp, z), \frac{W_x^3}{A^2} \frac{\partial |u|^2}{\partial W_x} \right\rangle \\ \dot{F}_y &= -\frac{1}{2n_0\alpha} W_y^4 + \frac{2}{n_0\alpha} F_y^2 + \frac{\gamma^2}{\alpha} \left\langle n(\mathbf{r}_\perp, z), \frac{W_y^3}{A^2} \frac{\partial |u|^2}{\partial W_y} \right\rangle \end{aligned} \quad (108)$$

3.5 Derivatives

$$|u|^2 = \frac{A^2 W_x W_y}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \quad (109)$$

$$\frac{\partial |u|^2}{\partial A} = 2A \frac{W_x W_y}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \quad (110)$$

$$\begin{aligned} \frac{\partial |u|^2}{\partial X} &= \frac{A^2 W_x W_y}{\pi} (2W_x^2(x-X)) \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \\ \frac{\partial |u|^2}{\partial Y} &= \frac{A^2 W_x W_y}{\pi} (2W_y^2(y-Y)) \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \end{aligned} \quad (111)$$

$$\begin{aligned} \frac{\partial |u|^2}{\partial W_x} &= \left(\frac{A^2 W_y}{\pi} - \frac{A^2 W_x W_y}{\pi} (2W_x(x-X)^2) \right) \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \\ &= (1 - W_x (2W_x(x-X)^2)) \frac{A^2 W_y}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \\ &= (1 - 2W_x^2(x-X)^2) \frac{A^2 W_y}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \end{aligned} \quad (112)$$

$$\begin{aligned}
\frac{\partial |u|^2}{\partial W_y} &= \left(\frac{A^2 W_x}{\pi} - \frac{A^2 W_y W_x}{\pi} (2W_y(y-Y)^2) \right) \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \\
&= (1 - W_y(2W_y(y-Y)^2)) \frac{A^2 W_x}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \\
&= (1 - 2W_y^2(y-Y)^2) \frac{A^2 W_x}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2))
\end{aligned} \tag{113}$$

$$V = \frac{1}{2A} \frac{\partial |u|^2}{\partial A} = \frac{W_x W_y}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \tag{114}$$

$$\begin{aligned}
\frac{W_x}{2A^2} \frac{\partial |u|^2}{\partial W_x} &= \frac{1}{2} (1 - 2W_x^2(x-X)^2) \frac{W_x W_y}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \\
&= \frac{1}{2} (1 - 2W_x^2(x-X)^2) V \\
\frac{W_y}{2A^2} \frac{\partial |u|^2}{\partial W_y} &= \frac{1}{2} (1 - 2W_y^2(y-Y)^2) \frac{W_x W_y}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \\
&= \frac{1}{2} (1 - 2W_y^2(y-Y)^2) V
\end{aligned} \tag{115}$$

$$\begin{aligned}
\frac{2}{A^2} \frac{\partial |u|^2}{\partial X} &= \frac{W_x W_y}{\pi} (4W_x^2(x-X)) \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \\
&= 4W_x^2(x-X) V \\
\frac{2}{A^2} \frac{\partial |u|^2}{\partial Y} &= \frac{W_x W_y}{\pi} (4W_y^2(y-Y)) \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \\
&= 4W_y^2(y-Y) V
\end{aligned} \tag{116}$$

$$\begin{aligned}
\frac{W_x^3}{A^2} \frac{\partial |u|^2}{\partial W_x} &= (1 - 2W_x^2(x-X)^2) \frac{W_x^3 W_y}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \\
&= W_x^2 (1 - 2W_x^2(x-X)^2) V \\
\frac{W_y^3}{A^2} \frac{\partial |u|^2}{\partial W_y} &= (1 - 2W_y^2(y-Y)^2) \frac{W_y^3 W_x}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \\
&= W_y^2 (1 - 2W_y^2(y-Y)^2) V
\end{aligned} \tag{117}$$

4 Calculating Statistics for Noise Terms: General

$$A(z) = A(0)$$

$$\dot{P} = \frac{1}{2n_0\alpha} (W_x^2 + W_y^2) - \frac{\gamma^2}{\alpha} \langle n(\mathbf{r}_\perp, z), V_P \rangle \quad (118)$$

$$\dot{X} = -2T_x$$

$$\dot{Y} = -2T_y \quad (119)$$

$$\dot{T}_x = -n_0\gamma^2 \langle n(\mathbf{r}_\perp, z), V_{T_x} \rangle$$

$$\dot{T}_y = -n_0\gamma^2 \langle n(\mathbf{r}_\perp, z), V_{T_y} \rangle \quad (120)$$

$$\dot{W}_x = 4F_x W_x$$

$$\dot{W}_y = 4F_y W_y \quad (121)$$

$$\dot{F}_x = -\frac{1}{2n_0\alpha} W_x^4 + \frac{2}{n_0\alpha} F_x^2 + \frac{\gamma^2}{\alpha} \langle n(\mathbf{r}_\perp, z), V_{F_x} \rangle$$

$$\dot{F}_y = -\frac{1}{2n_0\alpha} W_y^4 + \frac{2}{n_0\alpha} F_y^2 + \frac{\gamma^2}{\alpha} \langle n(\mathbf{r}_\perp, z), V_{F_y} \rangle \quad (122)$$

4.1 Modes

$$V = \frac{1}{2A} \frac{\partial |u|^2}{\partial A} = \frac{W_x W_y}{\pi} \exp(-(W_x^2(x-X)^2 + W_y^2(y-Y)^2)) \quad (123)$$

$$V_P = \frac{W_x}{2A^2} \frac{\partial |u|^2}{\partial W_x} + \frac{W_y}{2A^2} \frac{\partial |u|^2}{\partial W_y} + \frac{1}{2A} \frac{\partial |u|^2}{\partial A}$$

$$= \frac{1}{2} \left((1 - 2W_x^2(x-X)^2) + \frac{1}{2} (1 - 2W_y^2(y-Y)^2) + 1 \right) V \quad (124)$$

$$= (2 - W_x^2(x-X)^2 - W_y^2(y-Y)^2) V$$

$$V_{T_x} = 4W_x^2(x-X)V \quad (125)$$

$$V_{T_y} = 4W_y^2(y-Y)V \quad (126)$$

$$V_{F_x} = W_x^2 (1 - 2W_x^2(x - X)^2) V \quad (127)$$

$$V_{F_y} = W_y^2 (1 - 2W_y^2(y - Y)^2) V \quad (128)$$

It is natural to define the innerproduct as

$$\langle f, g \rangle = \text{Re} \left[\int \bar{f} g dt \right] \quad (129)$$

and then terms of the form

$$\langle n(\mathbf{r}, z), V(\mathbf{r}, \mathbf{p}(z)) \rangle = \text{Re} \left[\int n(\bar{\mathbf{r}}, z) V(\mathbf{r}, \mathbf{p}(z)) dt \right] \quad (130)$$

will have statistics

$$\mathbb{E}[\langle n(\mathbf{r}, z), V(\mathbf{r}, \mathbf{p}(z)) \rangle] = \text{Re} \left[\int \mathbb{E}[n(\bar{\mathbf{r}}, z)] V(\mathbf{r}, \mathbf{p}(z)) dt \right] \quad (131)$$

$$\mathbb{E}[\langle n(\mathbf{r}_1, z_1), V(\mathbf{r}_1, \mathbf{p}(z_1)) \rangle \langle n(\mathbf{r}_2, z_2), V(\mathbf{r}_2, \mathbf{p}(z_2)) \rangle] = \iint \iint \mathbb{E}[n(\mathbf{r}_1, z_1)n(\mathbf{r}_2, z_2)] V(\mathbf{r}_1, \mathbf{p}(z_1))V(\mathbf{r}_2, \mathbf{p}(z_2)) d\mathbf{r}_1 \quad (132)$$

4.2 Markov Approximation

If we assume the index is delta correlated in the propagation direction, than the correlations are easier to work with.

$$\mathbb{E}[n(\mathbf{r}_1, z_1)n(\mathbf{r}_2, z_2)] = C(|\mathbf{r}_1 - \mathbf{r}_2|)\delta(z_1 - z_2) = C(x_1 - x_2, y_1 - y_2)\delta(z_1 - z_2) \quad (133)$$

4.2.1 P

$$\begin{aligned} & \mathbb{E}[\langle n(\mathbf{r}_1, z_1), V_P(\mathbf{r}_1; W_x(z_1), W_y(z_1)) \rangle \langle n(\mathbf{r}_2, z_2), V(\mathbf{r}_2; W_x(z_2), W_y(z_2)) \rangle] \\ &= \iiint \iint C(x_1 - x_2, y_1 - y_2) V_P(x_1, y_1; W_x(z_1), W_y(z_1)) V_P(x_2, y_2; W_x(z_1), W_y(z_1)) d\mathbf{r}_1 d\mathbf{r}_2 \delta(z_1 - z_2) \end{aligned} \quad (134)$$

References

- [1] L.C. Andrews and R.L. Phillips. *Laser beam propagation through random media*. Press Monographs. SPIE Press, 2005.